Armaan Mittal

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EXPERIENCE

Software Engineer, Engagely.ai — June 2026 – Present

- Build a production **RAG assistant** that replaced manual **CDSCO regulatory lookup** for senior physicians, automating retrieval across a 2,000+ document corpus by **developing and consuming LLM APIs** (**Azure OpenAI**, **Cognitive Search**).
- Own **backend services and REST APIs** end-to-end (**Python**, **FastAPI**, **Celery**, **Redis**, **RabbitMQ**) on a **multi-tenant SaaS** platform, shipping from design through production deployment for enterprise clients (**Cipla**, **ART**).
- Design a **multi-agent orchestration system** with role-specialized agents and tool-calling to automate client workflows; build **Next.js** interfaces for internal tooling.
- Debug and resolve **production incidents** via **root-cause analysis**; build **load testing** (**1000 virtual users**) and implement **SSO/OIDC** authentication for secure multi-tenant access.

Research Intern (Part-time), Samsung R&D Institute Bangalore — Feb 2025 – Nov 2025

- Built low-latency **NLP** modules (**gibberish detection**, **PII** masking) for **Bixby** across 12 languages, achieving **93.4%** multilabel accuracy; built an **MCP server and client** to expose the tooling.
- **Deployed and monitored** models in production via **CI pipelines, Docker, and Kubernetes**; reduced end-to-end response latency by **15%** through **quantization** and cache-aware preprocessing.

Student Researcher, CoEDSAI, TIET Patiala — Jan 2026 – May 2026

- Built a **3D U-Net** pipeline for voxel-wise breast tumour segmentation on MRI volumes (PyTorch, MONAI), achieving **85.94% Dice** using patch-based training and a combined Dice + BCE loss to handle severe class imbalance.
- Co-developed **PRISM-Home** (paper under review, *Expert Systems With Applications*), an interpretable smart-home LLM system with hypothesis scoring, dry-run simulation, and an evidence-ledger verifier; achieved **0.91 grounded-clause precision** across 780 benchmark samples.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, Go, C++, SQL
AI & LLM: LLM APIs (OpenAI, Azure OpenAI), RAG, Azure Cognitive Search, Multi-Agent Systems, Tool Calling, MCP, ONNX Runtime, PyTorch
Integration & Deployment: FastAPI, REST APIs, SSO/OIDC, Next.js, React, PostgreSQL, Redis, RabbitMQ, Docker, Kubernetes, Cloudflare Workers, AWS, GCP, CI/CD

EDUCATION

Thapar Institute of Engineering and Technology, Patiala — August 2022 – May 2026

- BE in Electronics and Computer Engineering with **minor in Conversational AI**.
- **Coursework:** Operating Systems, Computer Networks, Data Structures & Algorithms, Computer Architecture, Object-Oriented Programming, Database Management Systems, Machine Learning, Generative AI.

PROJECTS

FarmSaathi | Multilingual Voice Assistant for Farmers — farmsaathi.amittal.dev | 2026

- Built a voice-first assistant in **Hindi, Punjabi and English**: **Whisper** speech-to-text, **RAG** over **bge-m3** embeddings in **Cloudflare D1**, and a fallback chain of **LLMs** (Groq, Gemini, OpenRouter).
- Moved the leaf-disease (15 classes) and crop-recommendation models into the browser with **ONNX Runtime Web** (tf2onnx, skl2onnx); parity tests against Python found **0 top-class mismatches** over 2,200 crop rows and 300 leaf images.
- Serves read-aloud with **MMS-TTS** on **Modal** (scale-to-zero) with an on-device fallback; ships as a 130 KB **PWA**.
- **Technologies: TypeScript, React, Hono, Cloudflare Workers, D1, Workers AI, ONNX Runtime Web, Modal.**

EduSched | Multi-Tenant Class Rescheduling Platform — edusched.amittal.dev | 2026

- Rebuilt a college timetable platform as a **Hono REST API** on **Cloudflare Workers** with **Neon PostgreSQL** (15 tables, workspace-scoped tenancy, coordinator and teacher roles, invites, class codes).
- Every reschedule runs in one transaction that takes **Postgres advisory locks** in a fixed order (no deadlocks) before re-checking teacher, room and batch **clashes**; publishes per-batch **iCalendar** feeds.
- Hashed **HttpOnly sessions**, origin-checked **CSRF** defence, rate-limited sign-in; API tests on in-process **PGlite**.
- **Technologies: TypeScript, Hono, Cloudflare Workers, PostgreSQL (Neon), React, Vitest, Vercel.**

ACHIEVEMENTS

- **Accepted research paper (ASCML 2026)**: "Scalable Multilingual Content Validation: A Dataset-Free Approach to Gibberish Detection".
- Selected for **Amazon Summer School 2024** (national AI/ML program) and **National Finalist**, Meta Pragati Hackathon.
- **Head, Competitive Programming Contest** (Saturnalia Fest), led a national-level competition with 300 participants; ranked **362 and 535** of 20,000+ on CodeChef [armaanmittal].
- Mentored 3K+ students in Data Structures & Algorithms via recorded modules and live sessions.